



Name : Sameer S
Reg no : 727723EUCB049

Name : Sameer S

Reg no : 727723EUCB049

Project Overview

The Budget Management System is a web-based application designed to help users plan and monitor their personal finances effectively.

It allows users to create custom budget categories, assign allocated amounts, and track expenses to ensure better financial control.

Key Highlights:

- Developed using React.js for a responsive and intuitive user interface.
 - Integrated with a Spring Boot backend via REST APIs for smooth communication and data handling.
 - Provides real-time budget summaries showing allocated, spent, and remaining amounts.
 - Supports adding, viewing, and deleting budget categories with instant UI updates.
- 

Project Objectives

The primary objective of the Budget Management System is to provide users with an efficient and user-friendly platform to plan and monitor their finances.

Specific Objectives:

- *Simplify Budget Creation:* Allow users to easily add and manage budget categories with names, allocated amounts, and descriptions.
 - *Enable Expense Tracking:* Display real-time summaries showing total allocation, spending, and remaining balances.
 - *Ensure Data Accuracy:* Validate all inputs and maintain consistent communication between the frontend and backend.
 - *Enhance User Experience:* Offer a clean, responsive, and intuitive interface built with React.js.
 - *Promote Efficient Management:* Provide quick insights into financial distribution and prevent overspending.
- 

Tech Stack

Frontend:

- React.js – Used to build a dynamic, responsive, and component-based user interface.
- Handles form validations, API integration, and real-time UI updates.

Backend:

- Spring Boot – Manages RESTful APIs for category management and budget summary.
- Provides secure, efficient data processing and connection to the database layer.

Communication:

- REST API – Facilitates smooth interaction between the frontend and backend using HTTP methods (GET, POST, DELETE).

Additional Tools:

- Axios for API calls in React
- Maven for dependency management in Spring Boot

Category Management Features

- Add new categories with name, allocated amount, and description
- Validate all input fields before submission
- View all existing categories with real-time updates
- Delete categories instantly with UI refresh
- Integrated with backend APIs:
 - POST /api/categories → Add
 - GET /api/categories → View
 - DELETE /api/categories/{id} → Delete

Form Representation

SKCET

React App

Presentation - Budget Manager

+

8081-baeadceaeaafcbebabbbfbbceef.premiumproject.examly.io

Personal Budget Tracker

Add New Category

Category Name *

Enter category name

Allocated Amount *

0.00

Description

Optional description

Add Category

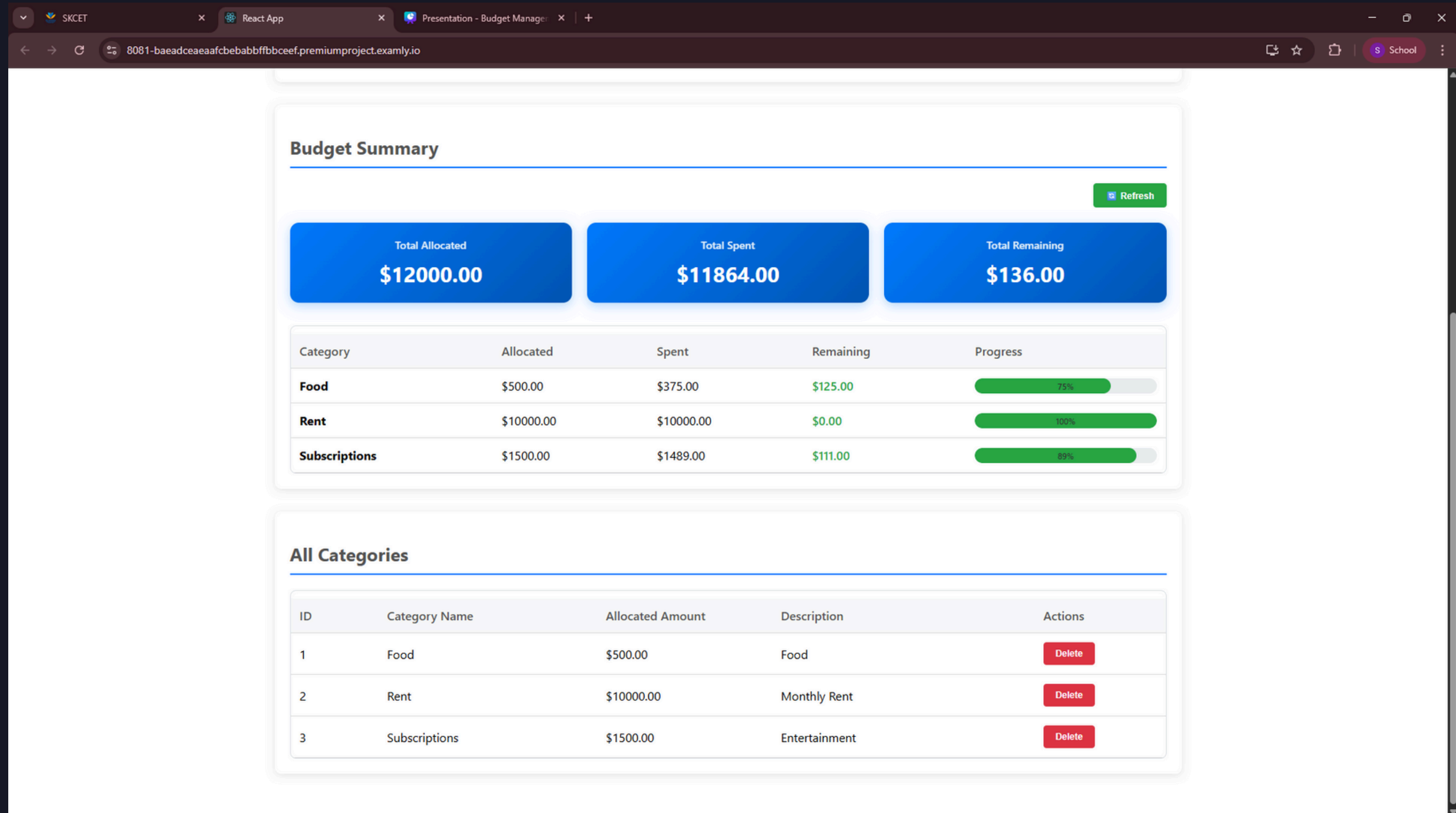
Budget Summary

No budget data yet. Add some categories to see your budget summary!

All Categories

No categories yet. Add your first category above!

Budget List View



Budget Summary Insights

- The Budget Summary module provides users with a clear overview of their financial status across all categories. It summarizes how funds are allocated, spent, and what remains.
- **Key Highlights:**
 - Displays each category's Allocated Amount, Spent Amount, and Remaining Balance.
 - Fetches live data from the backend using GET /api/categories/summary.
 - Automatically updates after any category addition or deletion.
 - Helps users identify overspending or underutilized budgets easily.
 - Presented through a clean and organized UI for quick financial understanding.
 - Lays the foundation for future features like expense tracking or graphical summaries.

API Integration

- The frontend and backend communicate seamlessly through RESTful APIs, ensuring real-time data synchronization and smooth user interactions.
- Key Endpoints:
 - POST /api/categories → Add a new budget category
 - GET /api/categories → Retrieve all categories
 - GET /api/categories/summary → Fetch budget summary details
 - DELETE /api/categories/{id} → Delete a category by ID
- Integration Highlights:
 - Implemented using Axios in React for efficient HTTP requests
 - Ensures instant UI updates after every API response

Key Strengths

- **Responsive Interface:** Built with React.js for smooth and fast user interaction.
- **Real-Time Updates:** Instant synchronization after every add or delete action.
- **Seamless API Integration:** Reliable communication between frontend and backend via REST APIs.
- **Clean and Intuitive Design:** User-friendly layout with simple navigation and validations.
- **Scalable Architecture:** Modular React components and maintainable Spring Boot backend.

Visual Representations

Personal Budget Tracker

Add New Category

Category Name *

Allocated Amount *

Description

Add Category

Budget Summary

No budget data yet. Add some categories to see your budget summary!

All Categories

No categories yet. Add your first category above!

Visual Representations

